Multimodal Fake News & Image Tampering Diagnostic Report

Generated on 2026-11-03 10:56:20 IST | Project Forensic Audit
Analysis performed at 2026-11-03 10:47:45 IST

1. Executive Summary & Verdict

Final Classification Verdict	**Likely Fake**
Overall Multimodal Risk Score	0.6668 (0.0=Real, 1.0=Fake)
Prediction Confidence	74.29%
Processing Time	1.03 seconds

This submission was classified as likely fake / misleading, with moderate confidence (74.3%). The article's text contributed most to this verdict (92% of the fused score), primarily because text model attribution flagged these tokens as most fake-indicative: claims, images, reportedly. Additional supporting evidence: error level analysis found no strong splicing/re-compression signal (anomaly score 0.068).

2. Evidence Summary

• Text model attribution flagged these tokens as most fake-indicative: claims, images, reportedly.
• Error Level Analysis found no strong splicing/re-compression signal (anomaly score 0.068).
• Noise-inconsistency analysis found unusually uneven noise levels across the image, a pattern sometimes associated with compositing.
• Edge-density analysis found an uneven distribution across image quadrants, consistent with a possible splice boundary.
• Simulated propagation graph shows an elevated proportion of suspicious/bot-like spreader accounts (20%).

3. Model Results & Fusion Weights

Modality	Model Engine	Risk Score	Weight	Relative Impact
Text Module	TF-IDF / BERT Transformer	0.8784	0.70	92.2%
Image Module	ResNet50 / ELA / pHash	0.0679	0.20	2.0%
Graph Module	PyG Graph Attention (GAT)	0.3834	0.10	5.8%

4. Analyzed Article Content

"A team of researchers has reportedly discovered unusual structures and unexplained energy patterns near Egypt's Great Pyramid of Giza, leading to renewed claims that extraterrestrial technology may have assisted ancient Egyptians in constructing the massive monuments. According to the report, satellite images reveal unusual geometric formations surrounding the pyramid that researchers believe coul..."

5. Text Explanation — Top Contributing Tokens

SHAP/coefficient attribution identified the following tokens as most influential in the text-model verdict: **claims, images, technology, reportedly, report, egypt, believe, according**.

6. Image Explanation

Error Level Analysis showed no strong splicing signal (anomaly score 0.0679). Perceptual hash: `c44438381c8142a6`. A GradCAM attention heatmap localizing the regions most influential to the ResNet-50 verdict was generated for this image (see dashboard for the visual overlay).

Forensic observations: noise levels vary unusually across regions of the image (coefficient of variation 1.00), a pattern sometimes associated with compositing content from a different source; edge density is unevenly distributed across the image quadrants (imbalance ratio 4.3x), which can indicate a localized splice boundary.

7. Propagation Analysis Summary

Simulated propagation graph: 45 nodes, 146 cascade edges, 5 detected communities, diffusion depth 5, 20% suspicious/bot-like accounts. Graph Attention Network propagation risk score: 0.3834.

8. Fusion Analysis

The final verdict is a weighted linear combination of the unimodal risk scores (see Section 3), calibrated via Platt scaling fit on the project's labeled demo corpus. Any modality not submitted for this analysis defaults to a neutral 0.5 score rather than being fabricated. The modality with the largest 'Relative Impact' in the table above contributed most to the final classification.

9. Final Decision & Recommendations

This content does not show strong indicators of fabrication based on the signals available to this system. As with any automated assessment, cross-verification against primary sources is still recommended before using this content for high-stakes decisions.